The Structure of Information Networks	**Reaction Paper**
CS/INFO 6850 Fall 2024	**Due 6pm Monday November 4, 2024**

The course includes a lot of material from recent research; this means that much of it exists mainly in the form of papers on the Web, and the existing literature raises a lot of interesting issues that have yet to be explored. As a way to get everyone thinking about the research issues underlying the course, and to prepare for the project, the next assignment is a short reaction paper of roughly 5 pages in length. You can work in groups of up to 4 people on the paper. The reaction paper is a good way to work on formulating a topic for the course project, and your group for the reaction paper can become your group for the project (which also can be done in groups of 4).

The reaction papers will be due at 6pm on November 4, and should be uploaded to Canvas. Even if you are working in a group of more than one, you should only hand in a single PDF for your group, with all of the members' names associated with it. (The system allows you to form a multi-person group that is associated with the submitted assignment.)

The reaction paper should be structured as follows. First, you should read at least two closely related papers relevant to a particular section of the course, neither of which is linked from the course home page. (You are certainly welcome to also use the papers linked from the course home page as resources, in addition to the two or more papers you focus on.) You should then write approximately 5 pages in which you address the following points:

- What is the main technical content of the papers?
- Which course topics is it related to, and why is it interesting in relation to these topics?
- What are some weakness of the papers, and how could they be improved?
- What are some promising further research questions in the direction of the papers, and how could they be pursued?

Reaction papers should not just be summaries of the papers you read; most of your text should be focused on synthesis of the underlying ideas, and your own perspective on the papers. To make this concrete, you should make sure that you devote much of the content to the last bullet above: directions for further research. In particular, the reaction paper should contain at least some amount of each of the following types of content:

- A proposal for a model or algorithm – potentially extending, varying, or improving something in the papers you've read – together with some mathematical analysis of it. (The analysis can consist of interesting mathematical definitions, basic properties of these definitions, and/or interesting worked-out examples; we don't expect new theorems, though you're welcome to include these if you make progress in this direction.)
- A test of a model or algorithm (either your own or something from one of the papers) on a dataset or on simulated data.

Of course, with only 5 pages, the work you do on these two points above will necessarily be preliminary; but even in preliminary form, it can be a good starting point for a project.